Daffodil International University

Department of Information and Communication Engineering
Faculty of Engineering

Capstone Project/Thesis/Internship Proposal Form

Date: 22.03.25 Serial No.: SP25010

Report Type: Capstone Project/Thesis/Internship

Sl	Student's ID	Student's Name	Mobile	CGPA	Capstone Project (ERP Registration)	Signature
01	221-50-083	MD. NAZMUL HOSEN BONDON	01782841067	3.84	N/A	
02	221-50-080	RIMPE DEB	01766816238	3.32	"	RIMPE
03	221-50-104	RABEYA AKTHER NEHA	01983002293	3.15	"	NEHA

Project Title:

IoT-based Smart Home System

Proposed Supervisor

Sl	Supervisor Name	Designation
01	Professor Dr. A.K.M. FAZLUL HAQUE	Professor
02	Engr. Md. Zahirul Islam	Assitant Professor
03	Dipto Biswas	Lecturer

Please contact the office to confirm your supervisor's name

Selected Supervisor

Sl	Supervisor Name	Designation
01	Prof. Dr. A.K.M Fazlul Haque.	Professor

20/4/25

Engr. Md. Zahirul Islam
Assistant Professor
Department of ICE

20/4/25

Dipto Biswas
Lecturer
Department of ICE

Approved By

Md. Taslim Arefin
Associate Professor & Head
Department of ICE

CamScanner

Daffodil International University

Department of Information and Communication Engineering

Capstone Project Synopsis – Industry Expert Review

Serial No.	SP25010
Project Title	IoT-based Smart Home System
Semester	Spring 2025
Students (ID – Name)	221-50-083 – Md. Nazmul Hosen Bondon 221-50-080 – Rimpe Deb 221-50-104 – Rabeya Akther Neha
Supervisor	Prof. Dr. A.K.M. Fazlul Haque, Professor, Department of ICE

Synopsis

Problem	Household appliances are usually operated manually, which wastes energy and gives no remote monitoring or early warning of hazards such as gas leakage or fire.
Objective	To design and implement an IoT-based system that monitors home conditions and allows users to control appliances remotely and automatically.
Proposed Method	A microcontroller (e.g. ESP32) with temperature, motion, light and gas sensors and relay modules; sensor data sent over Wi-Fi to a cloud platform; a mobile/web dashboard for monitoring, control and alerts; rule-based automation for lighting and appliances.
Expected Outcome	A working prototype with remote appliance control, real-time monitoring, safety alerts and reduced energy consumption.

Industry Expert Review

Industry relevance	☐ High ☐ Moderate ☐ Low
Comments on the topic	
Suggested modifications (scope, tools, standards, data, testing)	
Recommendation	☐ Approved as proposed ☐ Approved with modifications ☐ Not recommended
Name	Ferdous Alam
Designation	CEO
Organization	Atova Technology
Signature	
Date of review	

Supervisor: _____ Head, Department of ICE: _____

Daffodil International University

Department of Information and Communication Engineering
Faculty of Engineering

Capstone Project/Thesis/Internship Proposal Form

Date: 19.03.25 Serial No.: SP25006

Report Type: Capstone Project/Thesis/Internship

Sl	Student's ID	Student's Name	Mobile	CGPA	Capstone Project (ERP Registration)	Signature
01	221-50-094	Aishi Biswas	01730191140	3.47	N/A	Aishi
02	221-50-099	Md. Al Mustakim	01927130052	3.29	"	Mustakim
03	221-50-097	Fujilaturnnesa Chowdhury	01870534085	3.50	"	Fujilaturnnesa Chowdhury

Project Title: MediAI : A comprehensive AI system for medical diagnosis and treatment suggestions.

Proposed Supervisor

Sl	Supervisor Name	Designation
01	Md. Azizul Hakim	Lecturer
02	Dipto Biswas	Lecturer
03	Md. Saidur Rahman	Lecturer

Please contact the office to confirm your supervisor's name

Selected Supervisor

Sl	Supervisor Name	Designation
01	Md. Azizul Hakim	Lecturer

17/4/25

Engr. Md. Zahirul Islam
Assistant Professor
Department of ICE

20/4/25

Dipto Biswas
Lecturer
Department of ICE

Approved By

Md. Taslim Arefin
Associate Professor & Head
Department of ICE

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Department of Information and Communication Engineering

Capstone Project Synopsis – Industry Expert Review

Serial No.	SP25006
Project Title	MediAI: A Comprehensive AI System for Medical Diagnosis and Treatment Suggestions
Semester	Spring 2025
Students (ID – Name)	221-50-094 – Aishi Biswas 221-50-099 – Md. Al Mustakim 221-50-097 – Fujilatunnesa Chowdhury
Supervisor	Md. Azizul Hakim, Lecturer, Department of ICE

Synopsis

Problem	Online health information is often unreliable, and patients with low health literacy misunderstand prescriptions and make unsafe self-guided decisions.
Objective	To develop an AI-powered medical assistant that gives safe, explainable health information, explains prescriptions in plain language and supports diet planning, without replacing a doctor.
Proposed Method	Retrieval-Augmented Generation (RAG) on verified medical textbooks, clinical guidelines and drug references; OCR and medical entity extraction for prescriptions; a diet-planning module; a safety layer with confidence filtering and medical disclaimers.
Expected Outcome	A chatbot that answers medical queries with low hallucination, explains prescriptions accurately and improves patient understanding (final report: "AI-Powered Medical Assistant (Medi Chatbot)", December 2025).

Industry Expert Review

Industry relevance	☐ High ☐ Moderate ☐ Low
Comments on the topic	
Suggested modifications (scope, tools, standards, data, testing)	
Recommendation	☐ Approved as proposed ☐ Approved with modifications ☐ Not recommended
Name	Md. Hasan Ali
Designation	Senior AI Engineer
Organization	ACME AI
Signature	
Date of review	

Supervisor: _____ Head, Department of ICE: _____

Daffodil International University

Department of Information and Communication Engineering
Faculty of Engineering

Capstone Project/Thesis/Internship Proposal Form

Date: 19/03/25

Serial No.: SP25004

Report Type: Capstone Project/Thesis/Internship

Sl	Student's ID	Student's Name	Mobile	CGPA	Capstone Project (ERP Registration)	Signature
01	221-50-100	Abdiasis salad Ahmed	+8801821491675	2.93	N/A	Aziz
02	221-50-101	Abdihakim Mohamud Ahmed	+8801888517475	2.87	"	~~Hhhh~~
03					"	

Project Title: Development of IOT based smart ~~repaid~~ prepaid Energy ~~Em~~ meter.

Proposed Supervisor

Sl	Supervisor Name	Designation
01	Engr.md.ZahiruL ISLAM	Assistant professor
02	Md. Taslim Arefin	Associate Professor & Head
03	Dipto Biswas.	Lecturer

Please contact the office to confirm your supervisor's name

Selected Supervisor

Sl	Supervisor Name	Designation
01	Engr. Md. Zahirul Islam	Asst. Prof.

20/4/25

Engr. Md. Zahirul Islam
Assistant Professor
Department of ICE

Dipto Biswas 20/4/25

Dipto Biswas
Lecturer.
Department of ICE

Approved By

Md. Taslim Arefin
Associate Professor & Head
Department of ICE

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Daffodil International University

Department of Information and Communication Engineering

Capstone Project Synopsis – Industry Expert Review

Serial No.	SP25004
Project Title	Development of IoT-based Smart Prepaid Energy Meter
Semester	Spring 2025
Students (ID – Name)	221-50-100 – Abdiasis Salad Ahmed 221-50-101 – Abdihakim Mohamud Ahmed
Supervisor	Engr. Md. Zahirul Islam, Assistant Professor, Department of ICE

Synopsis

Problem	Conventional energy meters need manual reading and recharge, causing billing errors, delays and poor visibility of consumption for users.
Objective	To build a smart prepaid energy meter that measures consumption in real time, deducts balance automatically and lets users monitor and recharge online.
Proposed Method	Current and voltage sensing with a microcontroller; energy calculation and prepaid balance logic; relay-based automatic cut-off on zero balance; IoT connection to a cloud database and mobile/web app for recharge, usage history and low-balance alerts.
Expected Outcome	A prototype prepaid meter with real-time consumption monitoring, online recharge, automatic cut-off and alerts.

Industry Expert Review

Industry relevance	☐ High ☐ Moderate ☐ Low
Comments on the topic	
Suggested modifications (scope, tools, standards, data, testing)	
Recommendation	☐ Approved as proposed ☐ Approved with modifications ☐ Not recommended
Name	Kazi Mahbubul Alam
Designation	Deputy General Manager
Organization	Daffodil Online Ltd.
Signature	
Date of review	

Supervisor: _____ Head, Department of ICE: _____

Daffodil International University

Department of Information and Communication Engineering
Faculty of Engineering

Capstone Project/Thesis/Internship Proposal Form

Date: 19/05/25

Serial No.: SP25003

Report Type: Capstone Project/Thesis/Internship

Sl	Student's ID	Student's Name	Mobile	CGPA	Capstone Project (ERP Registration)	Signature
01	221-50-076	Ashraful Islam Ratul	01825532252	3.65	N/A	Ratul
02	221-50-077	Md. Sabbir Ahmed	01791111482	3.76	"	Sabbir
03	221-50-109	Md. Piash Islam	01301565464	3.83	"	Piash

Project Title: Developing an AI Integrated Smart Mentoring System (SMS) with Recommendation System Architecture.

Proposed Supervisor

Sl	Supervisor Name	Designation
01	Dipto Biswas	Lecturer
02	Md. Saidur Rahman Shawn	Lecturer
03	Md. Istakiak Adnan Palash	Lecturer

Please contact the office to confirm your supervisor's name

Selected Supervisor

Sl	Supervisor Name	Designation
01	Dipto Biswas	Lecturer

19/4/25

Engr. Md. Zahirul Islam
Assistant Professor
Department of ICE

Dipto Biswas 20/4/25

Dipto Biswas
Lecturer
Department of ICE

Approved By

Md. Taslim Arefin
Associate Professor & Head
Department of ICE

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Department of Information and Communication Engineering

Capstone Project Synopsis – Industry Expert Review

Serial No.	SP25003
Project Title	Developing an AI Integrated Smart Mentoring System (SMS) with Recommendation System Architecture
Semester	Spring 2025
Students (ID – Name)	221-50-076 – Ashraful Islam Ratul 221-50-077 – Md. Sabbir Ahmed 221-50-109 – Md. Piash Islam
Supervisor	Dipto Biswas, Lecturer, Department of ICE

Synopsis

Problem	Students often lack personalized guidance on courses, skills and career paths, and manual mentoring cannot scale to large batches.
Objective	To develop an AI-integrated mentoring platform that recommends learning resources, skills and mentors to each student based on their profile and progress.
Proposed Method	Student profile and performance data; recommendation engine combining content-based and collaborative filtering; web application for students and mentors; feedback loop to improve recommendations.
Expected Outcome	A working mentoring system that gives personalized recommendations and connects students with suitable mentors.

Industry Expert Review

Industry relevance	☐ High ☐ Moderate ☐ Low
Comments on the topic	
Suggested modifications (scope, tools, standards, data, testing)	
Recommendation	☐ Approved as proposed ☐ Approved with modifications ☐ Not recommended
Name	Irad Sakib
Designation	System Engineer
Organization	Nagad
Signature	
Date of review	

Supervisor: _____ Head, Department of ICE: _____

Daffodil International University

Department of Information and Communication Engineering
Faculty of Engineering

Capstone Project/Thesis/Internship Proposal Form

Date: 19/03/25 Serial No.: SP25 002

Report Type: Capstone Project/Thesis/Internship

Sl	Student's ID	Student's Name	Mobile	CGPA	Capstone Project (ERP Registration)	Signature
01	221-50-081	Methon Kumar Biswas	~~0186~~ 01838634685	3.47	N/A	Methon 19/03/25
02	221-50-082	Sethi Das	01864772191	3.37	"	Sethi Das 19.03.25
03	221-50-107 ~~25~~	Kazi Rahima Ahmed	01829004580	3.52	"	Rahima Ahmed 19/03/2025

Project Title: Implementing Smart Irrigation System: Real-Time Moisture Dependent Decision-making for Irrigation through AI-based Weather Station.

Proposed Supervisor

Sl	Supervisor Name	Designation
01	Dipto Biswas	Lecturer
02	Md. Azizul Hakim	Lecturer
03	Md. Saidur Rahman Shaown	Lecturer

Please contact the office to confirm your supervisor's name

Selected Supervisor

Sl	Supervisor Name	Designation
01	Dipto Biswas	Lecturer

19/4/25

Engr. Md. Zahirul Islam
Assistant Professor
Department of ICE

Dipto Biswas 20/4/25

Dipto Biswas
Lecturer
Department of ICE

Approved By

Md. Taslim Arefin
Associate Professor & Head
Department of ICE

CamScanner

Daffodil International University

Department of Information and Communication Engineering

Capstone Project Synopsis – Industry Expert Review

Serial No.	SP25002
Project Title	Implementing Smart Irrigation System: Real-Time Moisture Dependent Decision-making for Irrigation through AI-based Weather Station
Semester	Spring 2025
Students (ID – Name)	221-50-081 – Methon Kumar Biswas 221-50-082 – Sethi Das 221-50-107 – Kazi Rahima Ahmed
Supervisor	Dipto Biswas, Lecturer, Department of ICE

Synopsis

Problem	Fixed-schedule irrigation wastes water and energy and does not respond to actual soil moisture or expected rainfall.
Objective	To design an irrigation system that decides when and how much to irrigate from real-time soil moisture and weather station data.
Proposed Method	Soil moisture, temperature and humidity sensors with a microcontroller; a weather station and weather forecast data; an AI/ML model for irrigation decisions; automatic pump control and a dashboard for farmers.
Expected Outcome	A prototype that irrigates only when needed, reducing water use while keeping soil moisture at the required level.

Industry Expert Review

Industry relevance	☐ High ☐ Moderate ☐ Low
Comments on the topic	
Suggested modifications (scope, tools, standards, data, testing)	
Recommendation	☐ Approved as proposed ☐ Approved with modifications ☐ Not recommended
Name	Md. Mahbub Hasan Pavel
Designation	Senior Manager (Core Network Administration)
Organization	Amber IT Ltd.
Signature	
Date of review	

Supervisor: _____ Head, Department of ICE: _____